is produced by few strains of S. *aureus* which cleaves β – lactam ring of penicillin). Vancomycin is used against MRSA (Methicillin Resistant *Staphylococcus aureus*) strains.

Topical applications: For mild superficial lesions, topical applications of bacitracin or chlorhexidine is recommended.

Control measures: Proper sterilization of medical instruments must be done. Intake of antibiotics must be taken under proper medical advice. The detection of source & carriers among hospital staff, their isolation and treatment should be practiced.

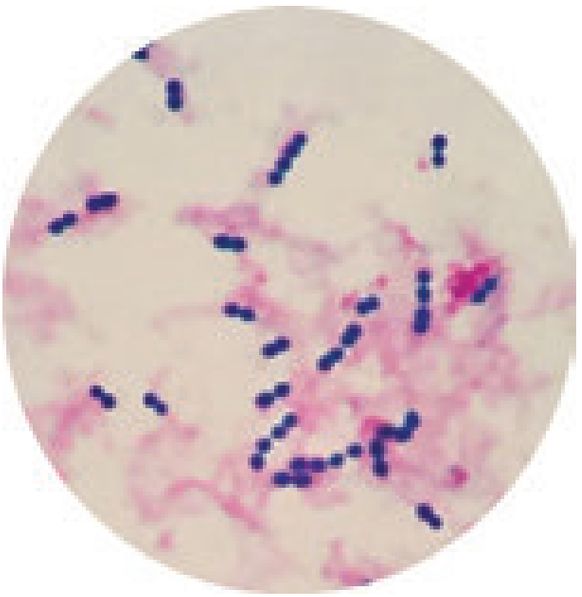


Figure 7.4: Gram staining of *Streptococcus pyogenes*

7.4 *Streptococcus Pyogenes* (Flesh eating Bacteria)

The genus *Streptococcus* includes a large and varied group of bacteria. They inhabit various sites, notably the upper respiratory tract. However, some species of which *Streptococcus* pyogenes is the most important and are highly pathogenic. The name *Streptococcus* is derived from **Greek** word **'Streptos'** which means twisted or coiled.

7.4.1 Morphology

- They are Gram positive, spherical or oval cocci and arranged in chains (0.6μm–1μm)
- They are non – motile, non – sporing. Some strains are capsulated (Figure 7.4).

7.4.2 Cultural Characteristics

- They are aerobe and facultative anaerobe. Optimum temperature is 37°C and pH is 7.4 to 7.6
- They grow only in media enriched with blood or serum. It is cultivated on blood agar. On blood agar, the colonies are small, circular, semitransparent, low convex, with an area of clear hemolysis around colonies (Figure 7.5).
- Crystal violet blood agar – a selective medium for *Streptococcus pyogenes.*



Year	Scientist	Contributions
1874	The odor Billroth	*Streptococci* were discovered
1884	Friedrich Julius Rosenbach	The cocci were isolated from human lesions and gave the name *Streptococcus pyogenes*

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